

Abdullah Emad Eldeen

AI & Embedded Systems Student | Machine Learning | TinyML | IoT

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PROFILE

Artificial Intelligence undergraduate at Menoufia University specializing in Intelligent Systems, with hands-on experience in Machine Learning, Deep Learning, Computer Vision, Embedded Systems, IoT, and TinyML. Experienced in building ML-based applications and integrating AI models with ESP32-class microcontrollers and sensors. Seeking AI/ML, Embedded AI, or IoT internship opportunities.

EDUCATION

Faculty of Artificial Intelligence, Menoufia University - B.Sc. in Artificial Intelligence, Intelligent Systems Department | 09/2023 - Present

Expected Graduation: 2027

TECHNICAL SKILLS

Programming	Python, C, C++
Machine Learning	Supervised Learning, Unsupervised Learning, Model Evaluation, Dimensionality Reduction, Feature Engineering
Deep Learning	Neural Networks, CNNs, Transfer Learning, Model Optimization
Computer Vision	OpenCV, MediaPipe, YOLO, Image Classification, Face Recognition
Embedded AI / IoT	ESP32, Arduino, Embedded Systems, Microcontrollers, TinyML, MQTT, I2S, Sensors
Frameworks / Tools	TensorFlow, PyTorch, NumPy, Pandas, Edge Impulse, TensorFlow Lite Micro, Arduino IDE, PlatformIO, ESP-IDF, FastAPI

PROJECTS

ESP32-S3 Keyword Spotting & TinyML

ESP32-S3 | INMP441 | TensorFlow Lite Micro | Edge Impulse | MFCC | I2S

- Developed an embedded keyword-spotting pipeline using an ESP32-S3 and INMP441 I2S microphone.
- Implemented 16 kHz audio acquisition and integrated microphone data with an on-device inference pipeline.
- Worked with MFCC-based audio preprocessing and quantized TensorFlow Lite Micro models for speech command classification.
- Debugged audio formatting, bit alignment, preprocessing consistency, and inference issues between training and embedded environments.

Smart Adaptive Environment Monitoring System

ESP32 | MQTT | Python | Machine Learning | IoT

- Developed an AI-enabled IoT monitoring system for real-time environmental sensing and adaptive control.
- Integrated ESP32 sensor acquisition with a Python processing and monitoring layer using MQTT.
- Applied PCA, Self-Organizing Maps (SOM), RBF-based modeling, Fuzzy Logic, and Reinforcement Learning techniques for analysis and decision-making.
- Built a live monitoring and control dashboard supporting hardware and simulation environments.

Real-Time ASL Alphabet Recognition

Python | TensorFlow/Keras | MediaPipe | Computer Vision

- Built a real-time American Sign Language alphabet recognition system that converts hand signs into text.
- Used MediaPipe for hand tracking and a custom-trained TensorFlow/Keras classification model for sign recognition.
- Integrated computer vision preprocessing with the classification pipeline for real-time inference.

Celebrity Face Recognition System

Python | FaceNet | ArcFace | FastAPI | HTML/CSS/JavaScript

- Developed a full-stack facial recognition application for celebrity identification.
- Integrated FaceNet and ArcFace-based embeddings for face representation and recognition.
- Built a FastAPI backend and web-based frontend for user interaction and inference.

TRAINING & COURSES

Generative AI Engineering with LLMs - IBM / Coursera (2026) | Deep Learning with PyTorch - IBM / Coursera | Fundamentals of TinyML - HarvardX / edX | TinyML2: Applications of TinyML - HarvardX / edX | Introduction to Embedded Machine Learning - Edge Impulse (2026) | Machine Learning Engineer Certification Preparation - Google Cloud / Coursera

LANGUAGES

Arabic - Native | English - Professional Working Proficiency | German - Basic